

Equity in Small Business Lending:

Predicting Loan Default and Auditing Fairness in SBA-Guaranteed Loans

Project Phase 2: Progress Report

CPSC 5071 — Data Management for Data Science — Winter 2026

Team Members: Bruna Porto and Jack Lichwa

GitHub Repository: <https://github.com/CPSC-5071-SELECT-ALL-FROM-STRUGGLING/FairLend-SBA>

1. Problem Statement

Small businesses are fundamental to the U.S. economy. The U.S. Small Business Administration (SBA) supports these businesses by guaranteeing loans to reduce lender risk [1]. However, when a guaranteed loan defaults, the financial loss is shared between the lending bank and U.S. taxpayers. Between 1987 and 2014, approximately 17.2% of SBA-guaranteed loans were charged off [2]. For lenders, accurately predicting which loans will default is essential for managing portfolio risk, setting appropriate loan terms, and making sound approval decisions.

At the same time, research has documented persistent disparities in access to capital for minority-owned and women-owned businesses [3]. Federal Reserve survey data confirm that these gaps persist, with minority-owned firms more likely to be denied credit or discouraged from applying [4]. As financial institutions increasingly adopt data-driven decision-making tools, a critical question emerges: could predictive models used in lending inadvertently encode or amplify these existing disparities?

Update based on initial EDA: Our exploratory analysis has revealed several important patterns that refine our understanding of the problem. Default rates varied dramatically over time, spiking to approximately 43% for loans approved during 2006–2008 (the Great Recession period) before declining to under 5% for post-2010 approvals. We also observe significant variation across industry sectors, with Real Estate (28.7%), Finance/Insurance (28.4%), and Transportation (26.9%) exhibiting the highest default rates. Geographic variation is similarly pronounced: Florida leads at 27.2%, followed by states hardest hit by the housing crisis (GA, DC, NV, MI). Additionally, 22.5% of loans are missing NAICS industry codes, which will limit our ability to join demographic composition data for those records. These findings confirm that loan characteristics, industry, geography, and macroeconomic timing all matter for default prediction, and they underscore the importance of integrating economic indicators from FRED alongside loan-level features.

Our research addresses two questions:

- **Predictive:** Can we predict SBA loan default based on loan characteristics, business attributes, and macroeconomic conditions?
- **Inferential:** Do loan outcomes differ across industry-state segments with varying concentrations of women-owned and minority-owned businesses? If deployed, would a predictive model produce equitable error rates across demographic segments?

2. Data Preprocessing

2.1 Data Sources

We use three primary data sources and one validation source, as described in our Phase 1 proposal:

Source	Description	Key Variables	Size	Join Key
--------	-------------	---------------	------	----------

SBA Loan Data (Kaggle)	SBA-guaranteed loans, 1987–2014	Loan amount, term, NAICS, state, default status	899,164 loans × 27 columns	NAICS + State + Year
Census ABS (API)	Industry × State demographic composition	pct_women_owed, pct_minority_owed by sector	3.96M rows × 16 cols (3 years: 2002, 2007, 2012)	NAICS + State + Year
FRED (API)	Monthly macroeconomic indicators	Fed Funds Rate, Prime Rate, CPI, Unemployment	Monthly, 1987–2014	Year + Month
Fed SBCS (Validation)	Firm-level survey on financing outcomes	Owner demographics, approval/denial rates	~7,600 firms	Reference only

2.2 Data Profiling (SBA Dataset)

We profiled the SBA dataset and identified the following data quality issues:

#	Issue	Columns Affected	Rows Affected	Action
1	Missing target variable (MIS_Status)	MIS_Status	1,997 (0.2%)	Drop rows
2	Target has embedded spaces ('P I F')	MIS_Status	All PIF rows	Strip/recode
3	Currency stored as strings (\$60,000.00)	DisbursementGross, GrAppv, SBA_Appv, BalanceGross, ChgOffPrinGr	All rows	Strip \$, to float
4	Dates stored as strings	ApprovalDate, DisbursementDate, ChgOffDate	All rows	Parse to datetime
5	Missing NAICS codes (industry = 0)	NAICS	201,948 (22.5%)	Recode as Unknown
6	Dirty categorical: RevLineCr	RevLineCr	277,479 not Y/N	Recode to Y/N/Unknown
7	Dirty categorical: LowDoc	LowDoc	6,007 not Y/N	Recode to Y/N/Unknown
8	NewExist has 0 values (unknown)	NewExist	1,034 rows	Recode 0 as NaN
9	ApprovalFY '1976A' transition quarter	ApprovalFY	18 rows	Recode to 1976
10	Sparse pre-1987 data	ApprovalFY	~600 rows	Filter to 1987+

After applying initial cleaning (dropping missing MIS_Status, filtering to 1987+, stripping MIS_Status whitespace, and converting currency columns), our working dataset contains 886,649 loans with a 17.2% default rate.

2.3 Data Cleaning

The following cleaning steps have been performed or are planned:

- **Completed:**
 - **SBA Dataset:** Target variable recoded to binary (CHGOFF = 1, PIF = 0); 1,997 rows with missing MIS_Status dropped; currency columns (DisbursementGross, GrAppv, SBA_Appv, BalanceGross, ChgOffPrinGr) stripped of \$ and commas and converted to float; 2-digit NAICS sector codes extracted; dataset filtered to 1987–2014 approval years.
 - **Census Survey of Business Owners (2002, 2007, 2012):** The raw Census data presented significant consistency challenges. The categorizations for demographic groups (SEX, RACE_GROUP,

ETH_GROUP) contained shifting terminology across survey years (e.g., highly verbose strings like "Publicly held and other firms not classifiable by gender..."). We standardized these varied classifications into a clean, uniform "Unclassified" category to prevent duplicate row counting. Furthermore, we identified and corrected a structural omission in the 2002 dataset where gender breakdowns for "All firms" under the Race category were missing, ensuring accurate aggregations for minority ownership.

- **NAICS Standardization:** Because NAICS industry definitions are updated every five years, the 2002 and 2007 Census datasets utilized obsolete sector codes. We utilized official U.S. Census Bureau crosswalk tables to map all historical NAICS codes to the unified 2012 standard (NAICS_2012_REV).
- **FRED Macroeconomic Data:** Monthly time-series data for the Federal Funds Rate (FEDFUNDS), Bank Prime Loan Rate (MPRIME), Consumer Price Index (CPIAUCSL), and Unemployment Rate (UNRATE) from 1987 to 2014 were joined and processed. We engineered a Year-over-Year (YoY) Inflation feature based on the CPI to better capture economic volatility.
- **Planned:** Full date parsing for ApprovalDate, DisbursementDate, ChgOffDate; recoding dirty categorical variables (RevLineCr, LowDoc, NewExist); handling NAICS = 0 records for Census joins; outlier detection on continuous variables (loan amounts, term, employees). Census ABS raw data has been collected for all three survey years (2002, 2007, 2012) via Census API and Dropbox; aggregation into demographic_profile format (computing pct_women_owned and pct_minority_owned by NAICS sector, state, and year) is in progress. FRED API data (Federal Funds Rate, Prime Rate, CPI, Unemployment) has been collected and merged.

2.4 Data Integration

We designed a relational database schema to integrate all three data sources. The schema consists of four tables in SQLite:

- **loan** — core table (one row per SBA loan), primary key: loan_id
- **industry_sector** — NAICS lookup table, primary key: naics_2digit
- **demographic_profile** — Census ABS data, composite primary key: (naics_2digit, state, data_year)
- **economic_indicator** — FRED time series, composite primary key: (year, month)

Join strategy: Each loan record links to its industry sector via naics_2digit, to its demographic profile via the composite key (naics_2digit, state, approval_year), and to macroeconomic conditions at the time of approval via (approval_year, approval_month).

3. Preliminary Results of EDA

3.1 Descriptive Statistics

The cleaned SBA dataset contains 886,649 loans approved between 1987 and 2014. The overall default rate is 17.2% (152,403 charged off vs. 734,246 paid in full). Key descriptive statistics:

- Median gross approved amount: approximately \$50,000; distribution is heavily right-skewed
- Loan terms range from 0 to 480+ months, with the most common term being 5–7 years (288,000 loans)
- 51 states/territories represented; California has the most loans (130,619), followed by Texas (70,458) and New York (57,693)
- 25 unique 2-digit NAICS sectors; 22.5% of loans have missing industry codes
- 72% of loans are to existing businesses; 28% to new businesses

3.2 Key Findings from Visualizations

Finding 1: Default rates vary dramatically over time. Loans approved during 2006–2008 exhibit default rates of approximately 43%, coinciding with the Great Recession and housing crisis. Post-2010 loans show very low default rates (<5%), though this may partly reflect right-censoring (newer loans have had less time to default). This temporal pattern confirms that macroeconomic conditions at the time of loan approval are a strong predictor of default, supporting our decision to integrate FRED economic indicators.

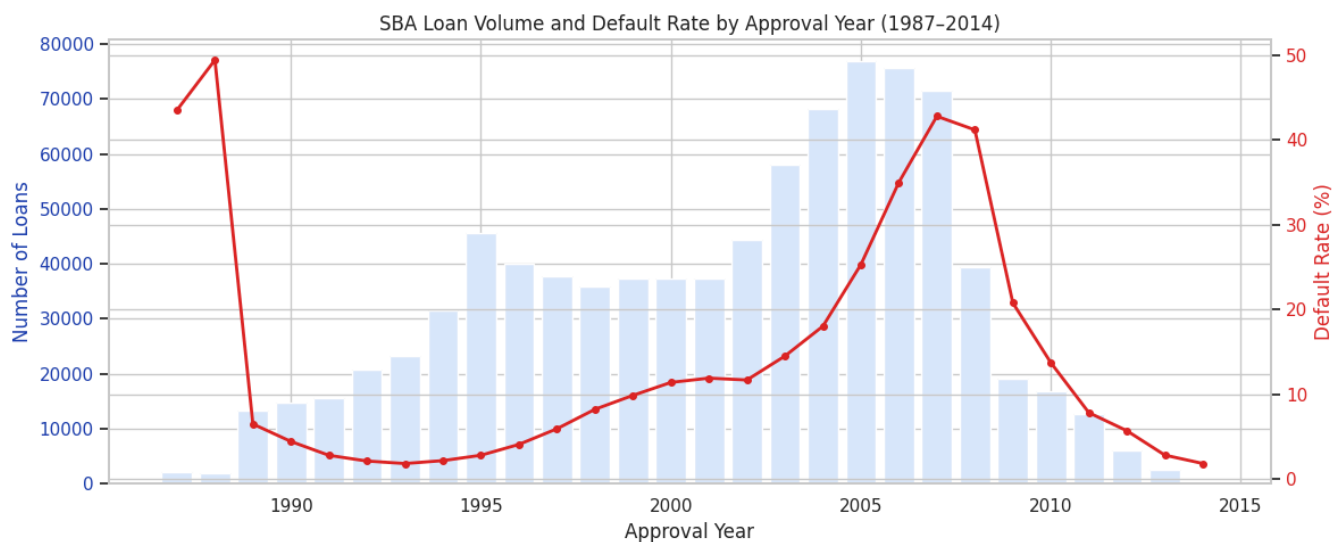


Figure 1: Default Rate Over Time

Finding 2: Defaulted loans skew toward smaller amounts. The distribution of gross approved amounts shows that defaulted loans are concentrated in the \$10,000–\$50,000 range, while paid-in-full loans have a higher proportion of larger loans approaching \$500,000. This suggests loan size may serve as a useful predictive feature, and that smaller loans carry proportionally higher risk.

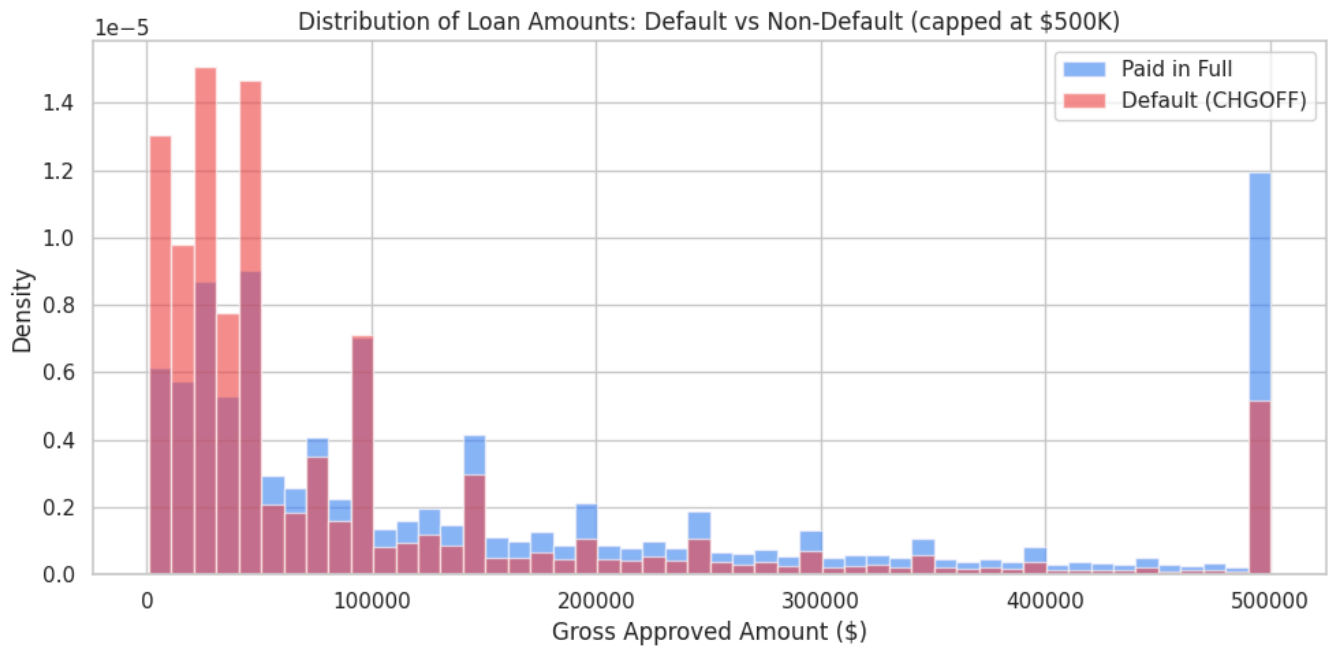


Figure 2: Loan Amount Distribution

Finding 3: Default rates differ substantially across industry sectors. Real Estate (28.7%), Finance/Insurance (28.4%), and Transportation (26.9%) have the highest default rates, while Healthcare (10.7%), Agriculture (9.6%), and Mining (9.3%) have the lowest. The Unknown NAICS category (22.5% of loans) has a notably low default rate (6.7%), warranting further investigation. These industry-level differences are directly relevant to our fairness analysis, as industry composition varies by demographic group.

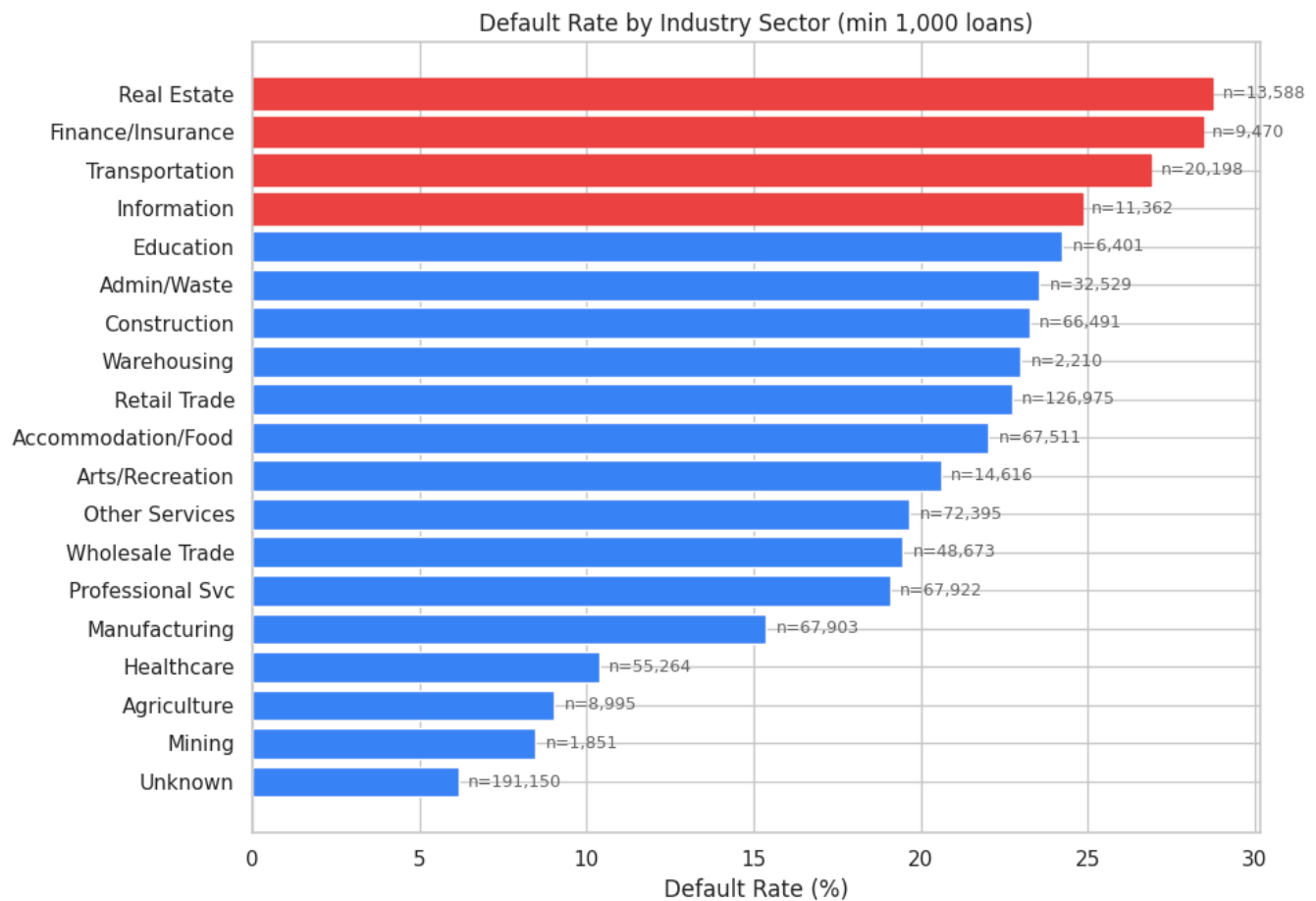


Figure 3: Default Rate by Industry

Finding 4: Geographic variation in defaults aligns with housing crisis patterns. Florida has the highest state-level default rate at 27.2% (41,032 loans), followed by Georgia, DC, Nevada, and Michigan. The lowest default rates are in Wyoming (5.5%), Montana (6.2%), and North Dakota (6.5%). This geographic variation is important for the fairness analysis because it intersects with the demographic composition of businesses in each state.

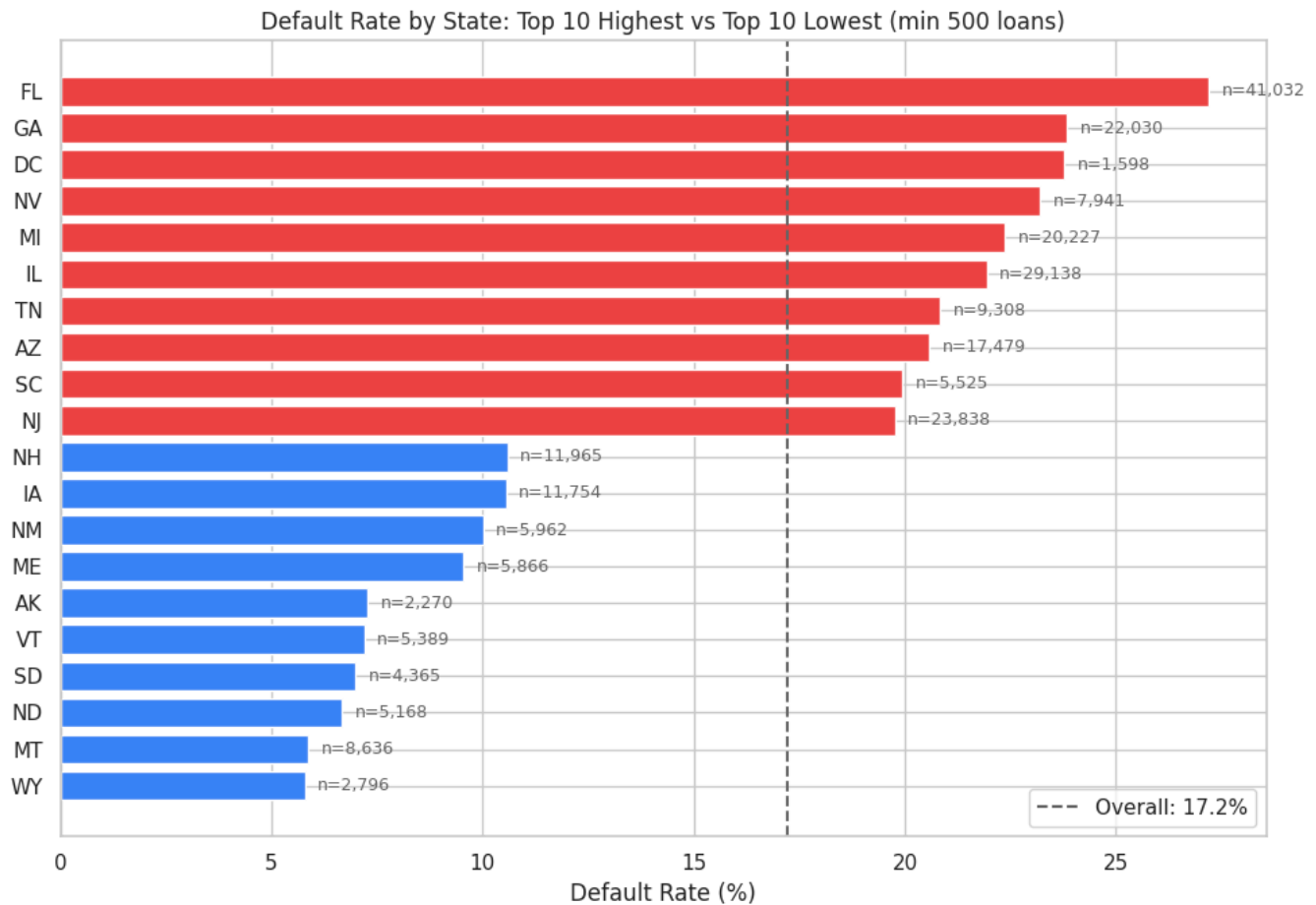


Figure 4: Default Rate by State

Finding 5: Shorter-term loans default at dramatically higher rates. Loans with terms of 3 years or less have a default rate of approximately 49%, declining steadily to about 3% for loans exceeding 20 years. This inverse relationship likely reflects the nature of the underlying assets: shorter-term loans are often unsecured working

capital, while longer-term loans are typically backed by real estate or equipment collateral.

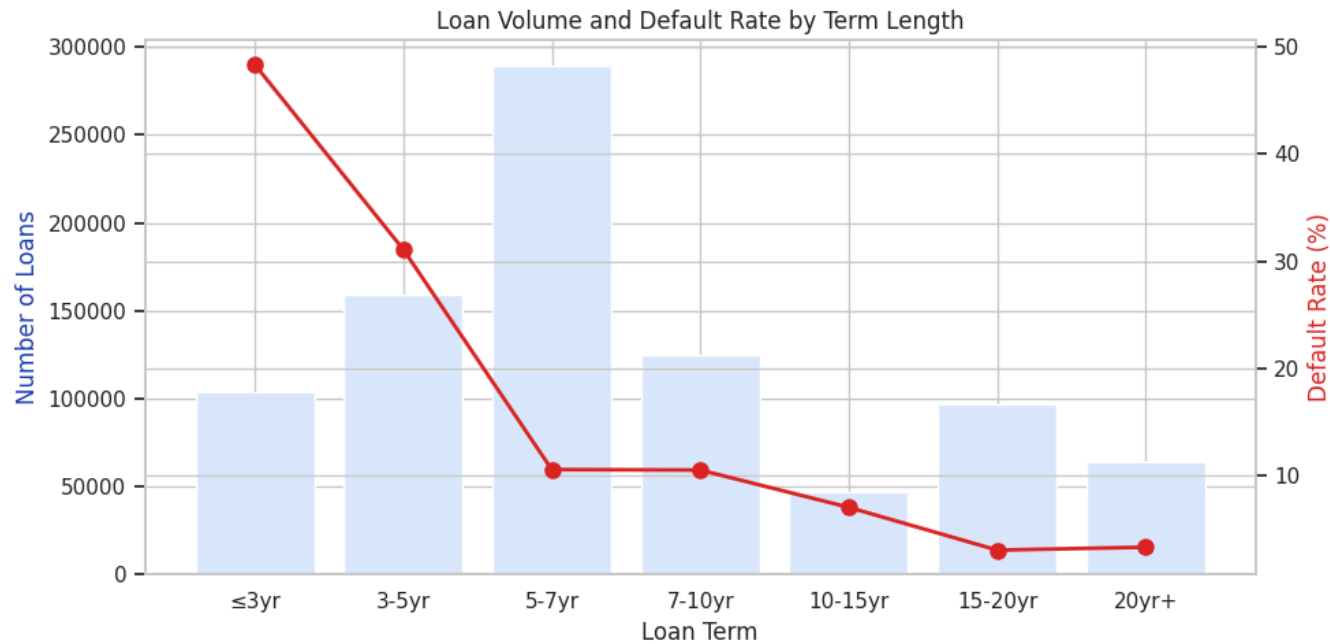


Figure 5: Default Rate by Term Length

Finding 6: New businesses have a modestly higher default rate. New businesses default at 18.3% compared to 16.8% for existing businesses. While the difference is relatively small (1.5 percentage points), it is meaningful given the large sample sizes (635,740 existing vs 249,765 new).

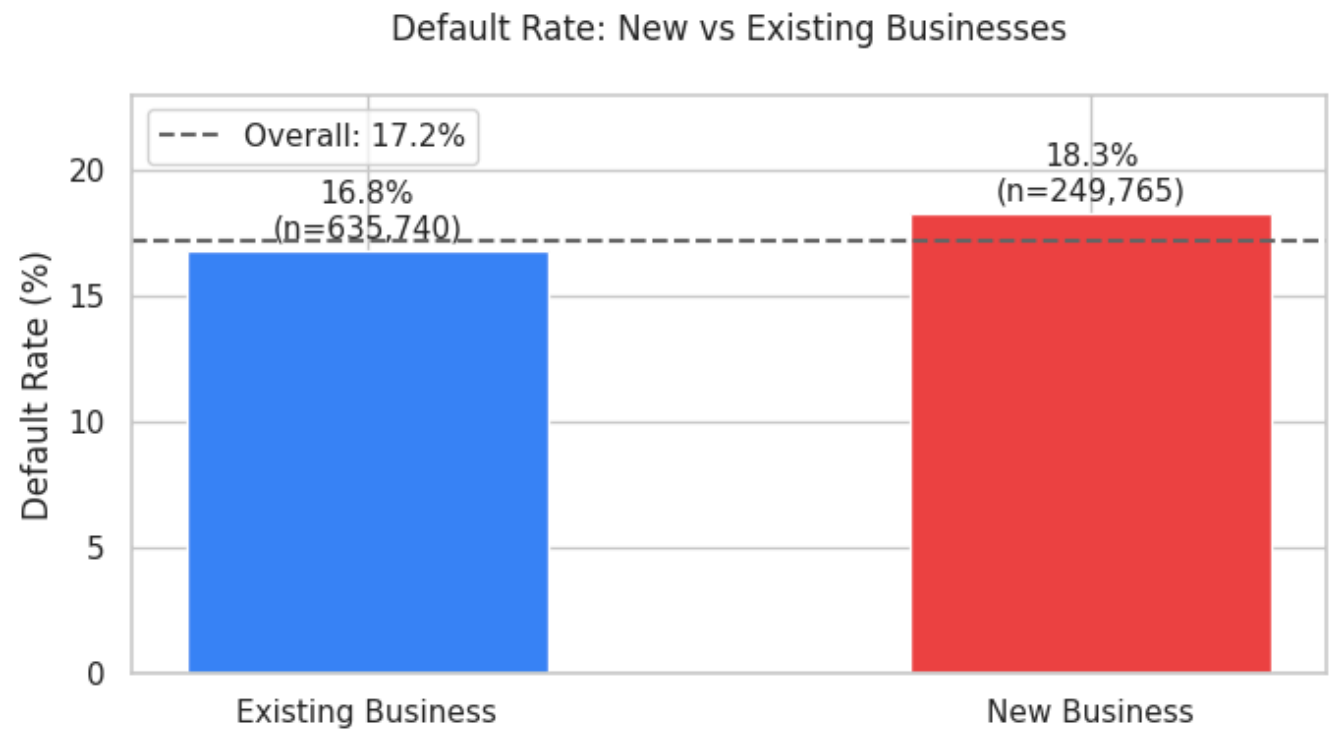


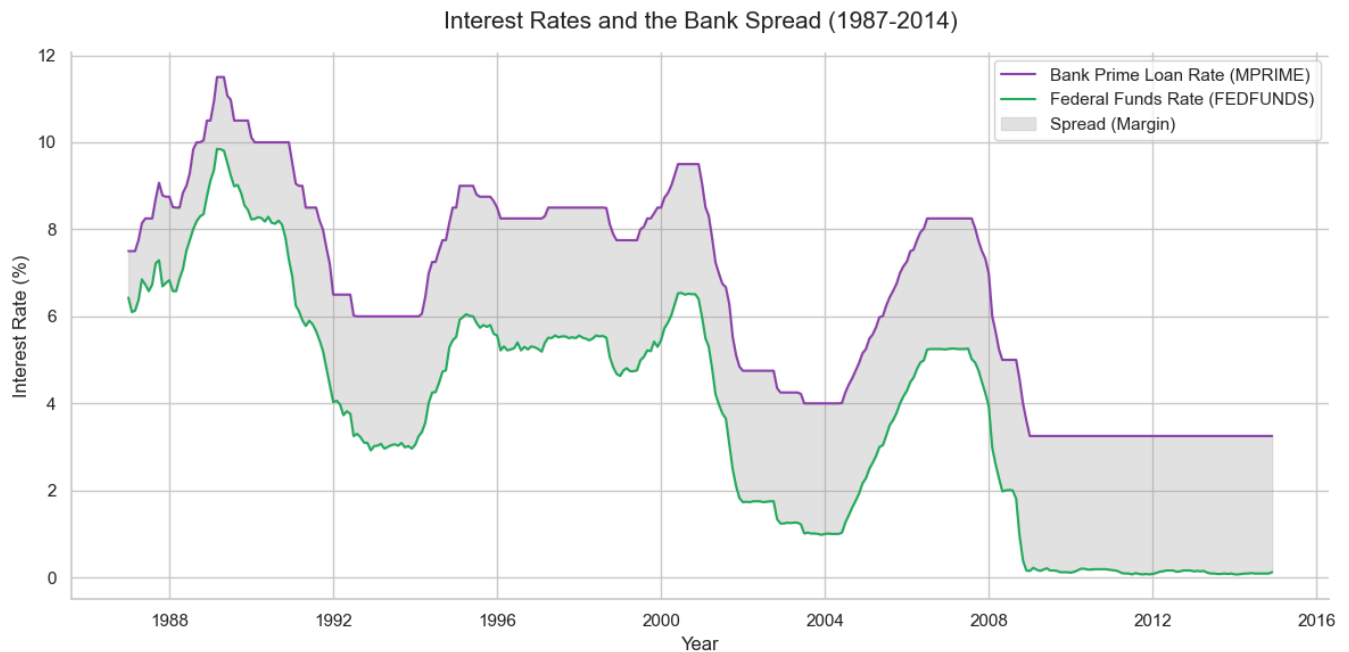
Figure 6: New vs Existing Business Default Rate

3.3 Macroeconomic Trends (FRED Data)

To contextualize the dramatic temporal spikes in SBA loan defaults—particularly the 43% default rate observed for 2006–2008 approvals—we conducted an exploratory analysis of historical macroeconomic indicators.

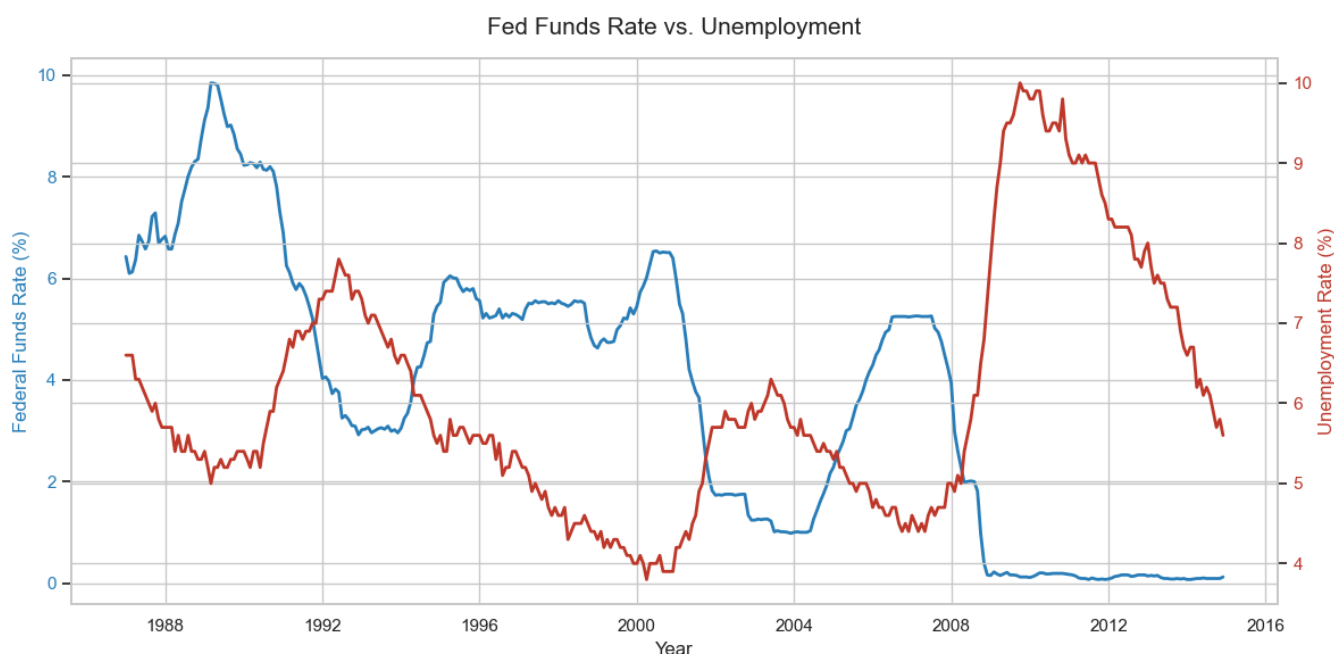
Finding 7: Interest rate drops and bank spreads trace the Great Recession.

An analysis of the Federal Funds Rate against the Bank Prime Loan Rate illustrates the credit environment for small businesses. While the spread (the margin between the Prime rate and the Fed Funds rate) remained relatively stable, the base rates plummeted dramatically following the onset of the 2008 financial crisis. This near-zero rate environment reflects emergency monetary policies enacted exactly when our SBA data shows default rates reaching their peak.



Finding 8: Unemployment correlates heavily with historical default spikes.

Plotting the National Unemployment Rate against YoY Inflation highlights the severe macroeconomic distress of the Great Recession. Unemployment surged past 9% between 2008 and 2010, closely mirroring the peak charge-off rates of the SBA loans approved immediately prior to the crash. This confirms that incorporating the unemployment rate at the time of loan maturity (or dynamically over the loan lifecycle) will be a critical predictor of loan survival.

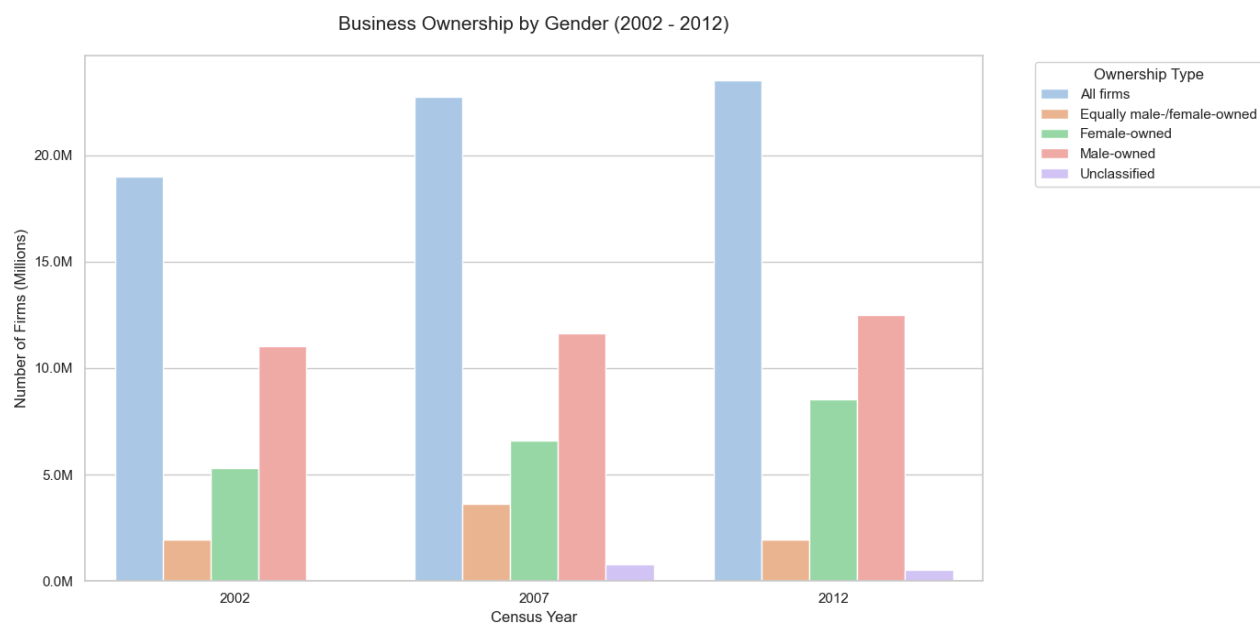


3.4 Demographic Business Ownership Trends (Census Data)

Because our fairness audit focuses on the equitable performance of predictive models across demographics, we evaluated the structural composition of U.S. businesses across the 2002, 2007, and 2012 Census periods.

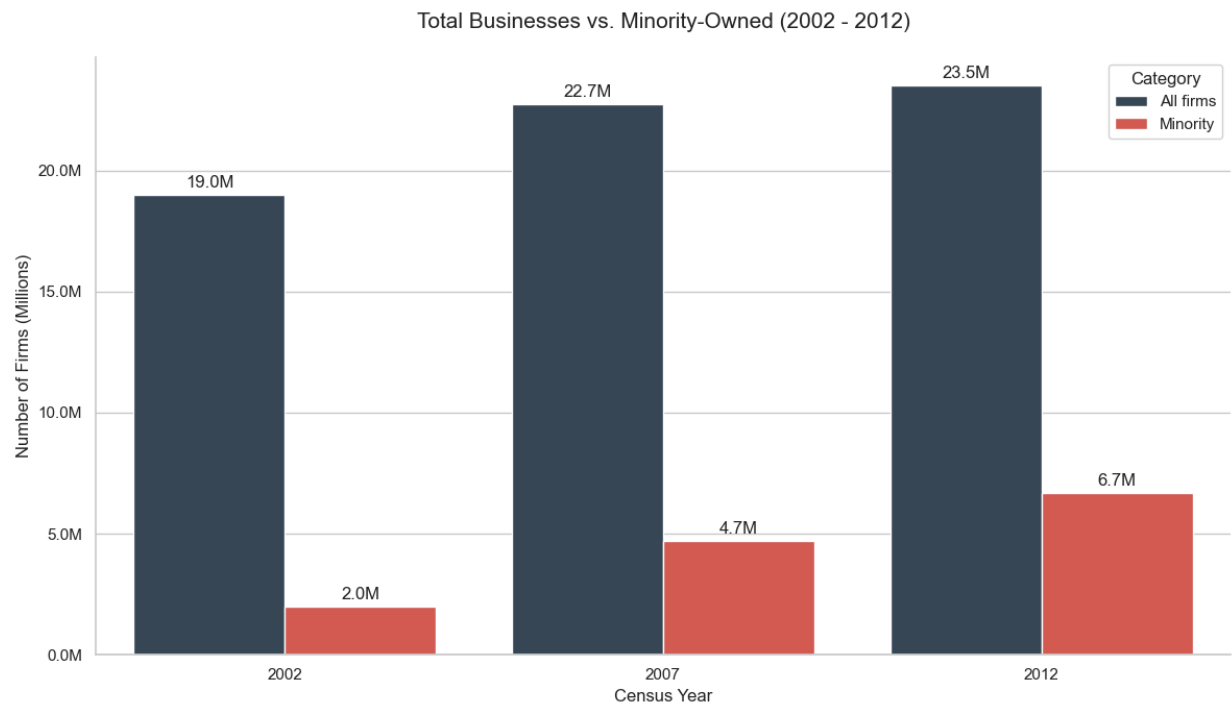
Finding 9: Total firm counts mask underlying shifts in demographic ownership.

Our deduplicated census aggregation shows steady growth in the total number of U.S. firms, growing from approximately 23 million in 2002 to over 27 million by 2012. However, a breakdown by gender reveals evolving ownership patterns. While male-owned businesses remain the dominant category, female-owned and equally-owned (male/female) businesses have captured an increasing share of total firm growth over this 10-year span.



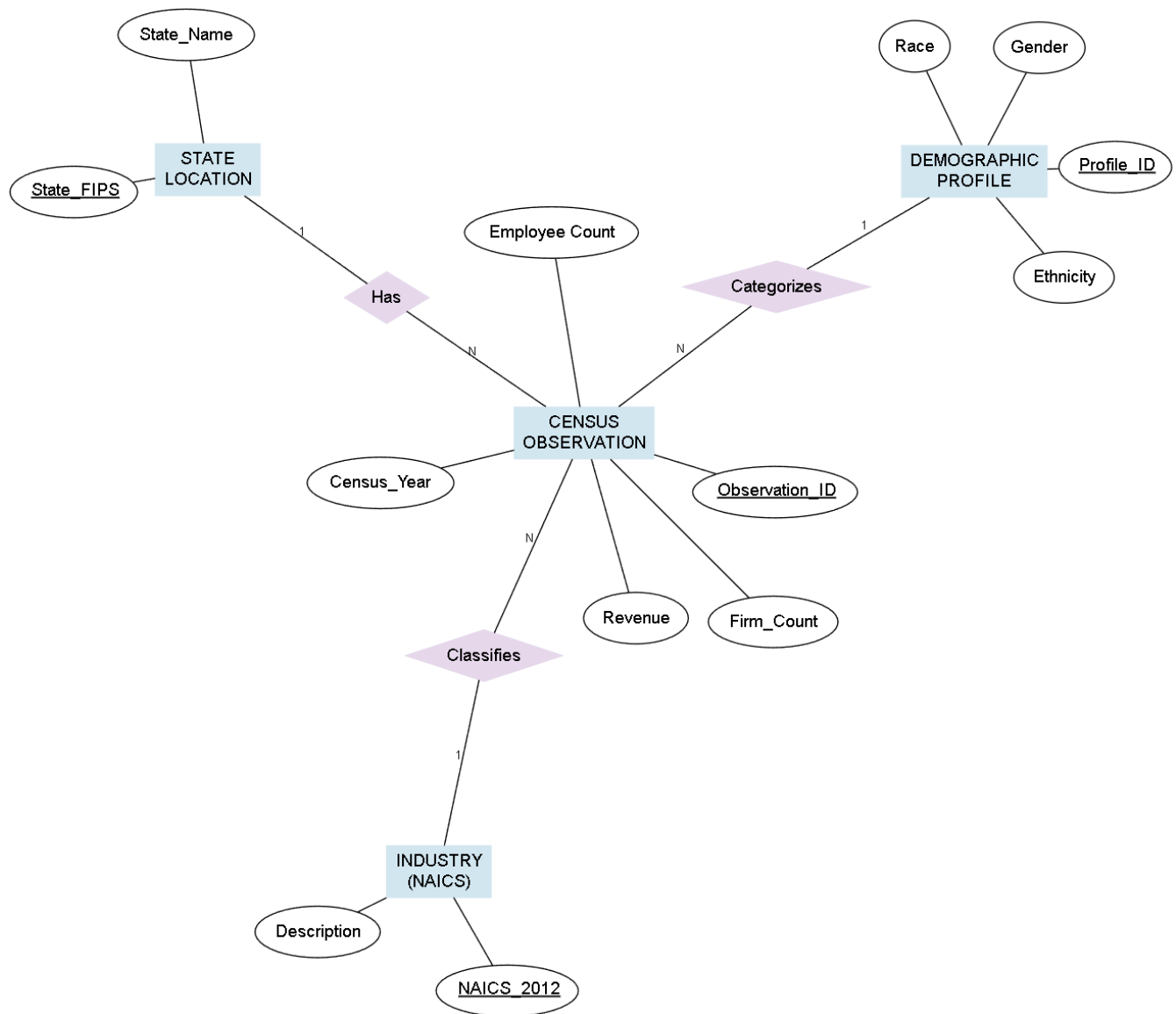
Finding 10: Minority-owned businesses are growing at an accelerated rate.

When isolating minority-owned firms against the total business population, we observe robust proportional growth. Despite the economic headwinds of the 2008 recession, the raw count of minority-owned businesses grew significantly from 2002 to 2012. Because minority ownership is not uniformly distributed across all industry sectors (NAICS) or geographic states, integrating these Census demographic proportions with the SBA loan data will allow us to test if models disproportionately penalize industry-state segments with high minority ownership concentrations.



4. Census Data Entity-Relationship Model

Based on your EDA data cleaning, and to address the slow loading and out-of-memory issues experienced during processing in Colab, the Census Survey of Business Owners is best modeled as a Star Schema. This structure features one central "Fact" entity surrounded by three "Dimension" entities.



5. Plan for Completion

5.1 Remaining Data Processing Tasks

- Aggregate Census ABS data into demographic_profile table (compute pct_women_owned, pct_minority_owned by NAICS sector × state × year from collected raw data)
- Clean and reshape FRED data for loading into economic_indicator table (data collected; formatting and monthly aggregation remaining)
- Full data integration: load all cleaned sources into SQLite, execute joins, and produce final modeling dataset
- Complete categorical recoding (RevLineCr, LowDoc, NewExist) and feature engineering
- Design relational model for Census data and load it to SQLite

5.2 Analysis and Modeling Plan

- **Logistic Regression (Jack):** Interpretable baseline model; stratified k-fold cross-validation; extract coefficients for inferential analysis of demographic covariates

- **Random Forest (Bruna):** Ensemble model with RandomizedSearchCV for hyperparameter tuning (n_estimators, max_depth, min_samples_split)
- **HistGradientBoosting (Bruna):** Gradient boosting model optimized for large datasets; handles missing values natively; tune learning_rate, max_iter, max_depth
- **Model comparison:** Evaluate all three models using AUC-ROC, precision, recall, F1-score; threshold analysis for optimal decision boundary

5.3 Evaluation and Validation Plan

- **Inferential analysis (Jack):** Test statistical significance of demographic composition covariates (pct_women_owned, pct_minority_owned) in the logistic regression after controlling for loan characteristics and economic indicators
- **Fairness audit (Bruna):** Disaggregate model error rates (FPR, FNR) across demographic subgroups using fairlearn MetricFrame; median-split industry×state segments into high vs. low women-ownership and minority-ownership groups; visualize disparities
- **Model inspection (Bruna):** Permutation importance and partial dependence plots to identify feature drivers and check for demographic proxy effects
- **SBCS validation (Jack):** Cross-check ecological-level patterns against firm-level findings from the Federal Reserve Small Business Credit Survey

6. References

- [1] U.S. Small Business Administration. “SBA Overview and History.” <https://www.sba.gov/about-sba>
- [2] Li, M., Mickel, A., and Taylor, S. “Should This Loan be Approved or Denied?: A Large Dataset with Class Assignment Guidelines.” *Journal of Statistics Education*, 26(1), 2018. Dataset: <https://www.kaggle.com/datasets/mirbektoktogaraev/should-this-loan-be-approved-or-denied>
- [3] Bates, T. and Robb, A. “Disparities in Capital Access between Minority and Non-Minority-Owned Businesses.” *The ANNALS of the American Academy of Political and Social Science*, 613(1), 2007.
- [4] Federal Reserve Banks. “2025 Report on Employer Firms: Findings from the 2024 Small Business Credit Survey.” <https://www.fedsmallbusiness.org/reports/survey/2025/2025-report-on-employer-firms>
- [5] U.S. Census Bureau. “Annual Business Survey (ABS).” <https://www.census.gov/data/developers/data-sets/abs.html>
- [6] Federal Reserve Bank of St. Louis. “FRED Economic Data.” <https://fred.stlouisfed.org/docs/api/fred/>
- [7] Barocas, S., Hardt, M., and Narayanan, A. *Fairness and Machine Learning: Limitations and Opportunities*. MIT Press, 2023.
- [8] Schweitzer, M. E. and Meyer, B. “Equal Access to Small-Business Credit.” Federal Reserve Bank of Cleveland, 2022.
- [9] U.S. Census Bureau. “Survey of Business Owners (SBO), 2012.” <https://www.census.gov/programs-surveys/sbo.html>